

SEQUENTIAL SYSTEMS

Phase II: Hardware Coercion

(Flip-Flop Conversions)

A Comprehensive Master Manual

Topics Covered:

The Philosophy of Hardware Synthesis

The Algorithmic 6-Step STC Methodology

The Master Excitation Arsenal

Synthesis: SR to JK

Synthesis: JK to D

Synthesis: D to T

Synthesis: SR to T

Rigorous K-Map Derivations & Vector Logic Diagrams

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Chapter 1

The Philosophy of Hardware Coercion

1.1 Introduction to Flip-Flop Conversion

In the realm of digital sequential design, we are frequently confronted with a fundamental engineering paradox: the hardware we physically possess does not match the logical behavior we mathematically require.

For instance, an engineering specification may demand a **Toggle (T) Flip-Flop** to construct a high-speed synchronous binary counter. However, the available inventory, or the constraints of the Application-Specific Integrated Circuit (ASIC) cell library, may only provide **Data (D) Flip-Flops** or **Set-Reset (SR) Flip-Flops**.

1.1.1 The Concept of Hardware Coercion

We cannot physically alter the silicon inside the microchip. An SR Flip-Flop will forever remain an SR Flip-Flop at the transistor level.

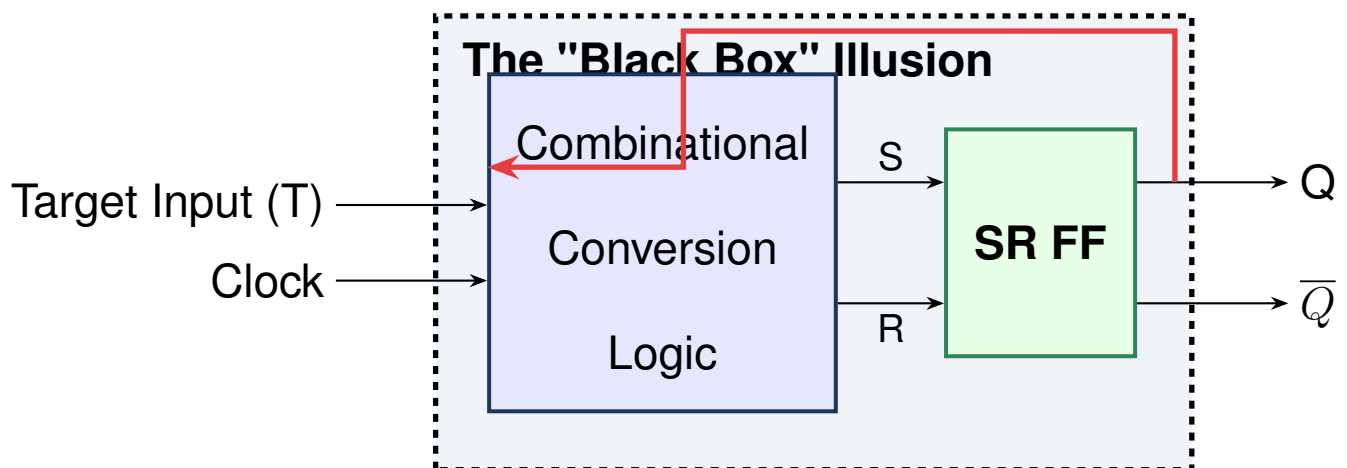
Instead, we employ a technique known as **Hardware Coercion** or **Flip-Flop Conversion**. We design an external layer of Combinational Logic (AND, OR, NOT, XOR gates) that wraps around the available flip-flop. This combinatorial shell intercepts the "Target" input signals, mathematically transforms them, and injects the resulting

"Source" excitation signals directly into the physical pins of the microchip.

1.2 The Illusion of the Black Box

The ultimate goal of hardware coercion is to create a perfect logical illusion.

If we wrap an SR flip-flop in the correct combinational logic, we can place a "Black Box" over the entire assembly. To the outside observer, the black box possesses only a single 'T' input pin and a clock. When the observer pulses $T = 1$, the box toggles perfectly. The observer believes they are interacting with a native T-Flip-Flop, completely unaware that a coerced SR-Flip-Flop is actually driving the core mechanics inside.



Chapter 2

The 6-Step STC Synthesis Methodology

2.1 Algorithmic Formalization

Hardware coercion cannot be achieved through intuition or trial-and-error. It requires a rigorous, mathematically sound, algorithmic approach. In the STC Sir syllabus framework, every single flip-flop conversion **MUST** adhere to the following absolute 6-Step procedure to guarantee maximum marks and zero logical faults.

2.1.1 Step 1: Define the Source and Target

Clearly state the available hardware (The Source) and the desired behavior (The Target). Write down the Characteristic Truth Table for the Target, and the Excitation Table for the Source.

2.1.2 Step 2: Construct the State Synthesis Table

This is a monolithic table that merges both sets of logic. It dictates exactly what inputs must be injected into the Source pins to satisfy the Target's demands for every possible

present state (Q_n).

2.1.3 Step 3: Extract the K-Map Parameters

Extract the columns corresponding to the Source Input pins (e.g., S and R). These will serve as the outputs of our combinational logic. The inputs to the K-Map will be the Target Inputs (e.g., J, K) and the Present State (Q_n).

2.1.4 Step 4: Execute Karnaugh Map Minimization

Draw rigorous Karnaugh Maps for every single Source input pin. Group the 1s, aggressively utilize "Don't Cares" (X) to maximize grouping size, and extract the highly minimized Sum of Products (SOP) Boolean expressions.

2.1.5 Step 5: Boolean Equation Verification

Analyze the resulting Boolean equations to ensure they are physically realizable with standard logic gates (AND, OR, NOT, XOR).

2.1.6 Step 6: Render the Physical Logic Schematic

Draw the final architectural circuit diagram. The Source Flip-Flop sits in the center. The combinational logic gates derived in Step 4 are placed to its left. Crucially, the feedback wires from Q and $\overline{Q}$ must be meticulously routed back into the combinational logic.

Chapter 3

The Arsenal: Excitation Tables

3.1 The Prerequisite for Synthesis

Before we can execute the 6-Step methodology, we must have absolute mastery over the Excitation Tables of the four core flip-flops. If you attempt a conversion without these tables memorized, the mathematics will immediately collapse.

3.1.1 The SR Flip-Flop (Set-Reset)

Q_n	Q_{n+1}	S (Set)	R (Reset)
0	0	0	X (Don't Care)
0	1	1	0
1	0	0	1
1	1	X (Don't Care)	0

3.1.2 The JK Flip-Flop (Universal)

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

3.1.3 The D Flip-Flop (Data / Delay)

The D Flip-Flop is beautifully transparent. The required input D is always exactly equal to the desired next state.

Q_n	Q_{n+1}	D (Data)
0	0	0
0	1	1
1	0	0
1	1	1

3.1.4 The T Flip-Flop (Toggle)

The T Flip-Flop requires a 1 to change state, and a 0 to maintain state.

Q_n	Q_{n+1}	T (Toggle)
0	0	0 (Hold)
0	1	1 (Toggle)
1	0	1 (Toggle)
1	1	0 (Hold)

Chapter 4

Synthesis I: SR to JK Conversion

4.1 Step 1 & 2: The State Synthesis Table

Target: JK Flip-Flop

Source Hardware: SR Flip-Flop

We must map the three independent variables (J , K , and Q_n) to determine the target next state (Q_{n+1}). Then, we consult the SR Excitation Table to find the required S and R voltages.

Target Parameters			Result	Source Needs	
J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

4.2 Step 3 & 4: Karnaugh Map Minimization

We now construct two 3-variable K-Maps. One map will solve for S , and the other will solve for R .

4.2.1 Deriving the Equation for 'S'

$$KQ_n$$

J	00	01	11	10
0	0	X	0	0
1	1	X	0	1

Minimization Logic: We group the 1 at cell 4 with the 1 at cell 6. The variable K changes (0 to 1), so it drops out. J is constant at 1. Q_n is constant at 0 (which means $\overline{Q_n}$).

Final Equation: $S = J \cdot \overline{Q_n}$

4.2.2 Deriving the Equation for 'R'

$$KQ_n$$

J	00	01	11	10
0	X	0	1	X
1	0	0	1	0

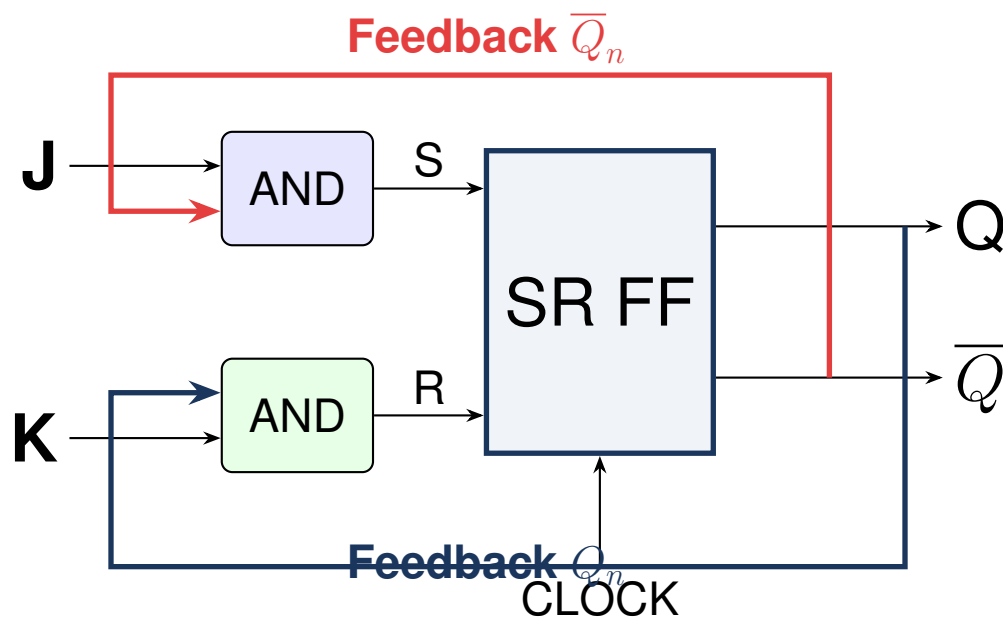
Minimization Logic: We group the 1 at cell 3 with the 1 at cell 7. The variable J changes (0 to 1), so it drops out. K is constant at 1. Q_n is constant at 1.

Final Equation: $R = K \cdot Q_n$

4.3 Step 5 & 6: Logical Implementation

The mathematical equations dictate that we require two 2-input AND gates.

- The first AND gate generates S by multiplying the external J input with the feedback wire from $\overline{Q}$.
- The second AND gate generates R by multiplying the external K input with the feedback wire from Q .



Chapter 5

Synthesis II: JK to D Conversion

5.1 Step 1 & 2: The State Synthesis Table

Target: D (Data) Flip-Flop

Source Hardware: JK Flip-Flop

This is a much simpler conversion because the D flip-flop only possesses a single external input (D). Therefore, our truth table will only evaluate two independent variables: D and Q_n .

Target		Result	Source Needs	
D	Q_n	Q_{n+1}	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

5.2 Step 3 & 4: Karnaugh Map Minimization

We construct two 2-variable K-Maps to solve for J and K.

5.2.1 Deriving the Equation for 'J'

	Q_n	
D	0	1
0	0	X
1	1	X

Minimization Logic: We group the 1 with the X horizontally. The variable Q_n drops out entirely.

Final Equation: $J = D$

5.2.2 Deriving the Equation for 'K'

	Q_n	
D	0	1
0	X	1
1	X	0

Minimization Logic: We group the 1 with the X horizontally on the top row ($D = 0$).

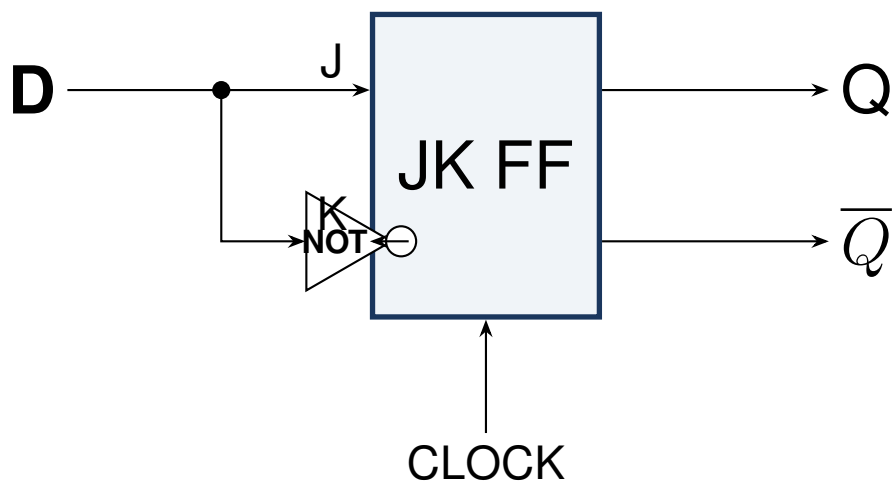
The variable Q_n drops out entirely.

Final Equation: $K = \overline{D}$

5.3 Step 5 & 6: Logical Implementation

The mathematics reveal an elegant solution. The source J input simply requires the raw D signal. The source K input requires the exact mathematical inverse of the D signal.

We only require a single NOT gate (Inverter) to achieve this hardware coercion. No feedback loops from Q or $\overline{Q}$ are necessary!



Chapter 6

Synthesis III: D to T Conversion

6.1 Step 1 & 2: The State Synthesis Table

Target: T (Toggle) Flip-Flop

Source Hardware: D (Data) Flip-Flop

We evaluate the inputs T and Q_n . Because the Source is a D Flip-Flop, the Excitation column is brilliantly simple: D must exactly equal Q_{n+1} .

Target		Result	Source Needs
T	Q_n	Q_{n+1}	D
0	0	0 (Hold)	0
0	1	1 (Hold)	1
1	0	1 (Toggle)	1
1	1	0 (Toggle)	0

6.2 Step 3 & 4: Karnaugh Map Minimization

6.2.1 Deriving the Equation for 'D'

	Q_n	
T	0	1
0	0	1
1	1	0

Minimization Logic: The 1s are located diagonally across the K-Map (at minterms 1 and 2). They cannot be grouped together. We must write the equation as a standard Sum of Products.

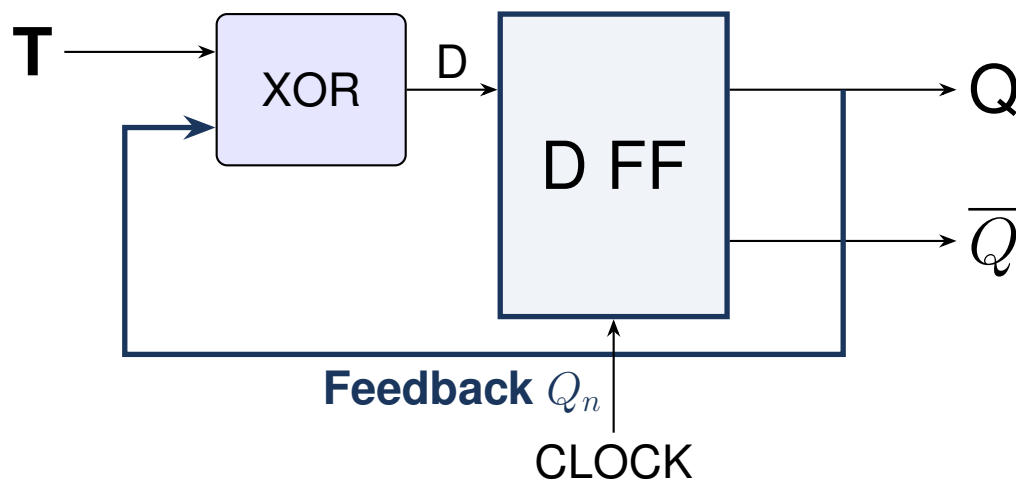
$$D = \overline{T}Q_n + T\overline{Q}_n \quad (6.1)$$

By recognizing the fundamental Boolean identities, we can compress this equation into an Exclusive-OR (XOR) operation:

$$\mathbf{D} = \mathbf{T} \oplus \mathbf{Q}_n \tag{6.2}$$

6.3 Step 5 & 6: Logical Implementation

The mathematics dictate that the Source D pin must be fed by an XOR gate. The XOR gate will receive the external T signal on one leg, and the feedback wire from the present state Q on the other leg.



Chapter 7

Synthesis IV: SR to T Conversion

7.1 The Ultimate Coercion Challenge

Converting an SR flip-flop directly into a T flip-flop represents the highest level of foundational synthesis, as it requires navigating the SR flip-flop's forbidden state while enforcing a strict toggling protocol.

7.1.1 State Synthesis Table

Notice how $T = 1$ forces the SR flip-flop to explicitly Set (1,0) when $Q = 0$, and explicitly Reset (0,1) when $Q = 1$. The "Don't Cares" only exist when $T = 0$ (the hold states).

Target		Result	Source Needs	
T	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	X	0
1	0	1	1	0
1	1	0	0	1

7.2 Karnaugh Map Minimization

7.2.1 Deriving the Equation for 'S'

	Q_n	
T	0	1
0	0	X
1	1	0

Minimization Logic: The 1 at minterm 2 cannot be grouped with the X at minterm 1 because they are diagonal. We isolate the 1.

Final Equation: $S = T \cdot \overline{Q_n}$

7.2.2 Deriving the Equation for 'R'

	Q_n	
T	0	1
0	X	0
1	0	1

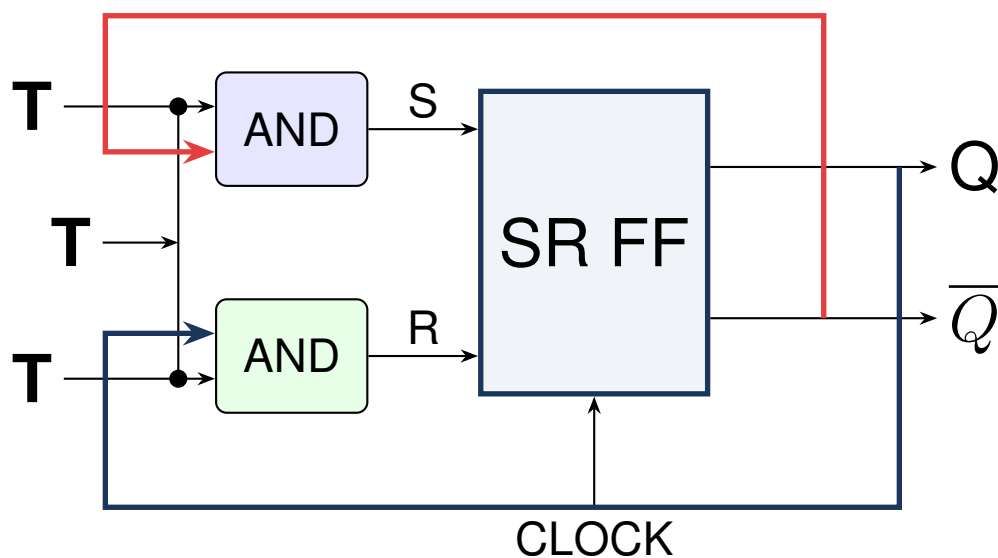
Minimization Logic: The 1 at minterm 3 cannot be grouped with the X at minterm 0.

We isolate the 1.

Final Equation: $R = T \cdot Q_n$

7.3 Final Logical Implementation

The equations dictate a remarkably elegant dual-feedback loop architecture. We require two AND gates. Both gates receive the external T toggle command. The Set pin is gated by the inverted $\overline{Q}$ feedback, and the Reset pin is gated by the normal Q feedback.



***** END OF PHASE II MASTER
MANUAL *****